

Md Hasibul Amin

Software Engineer — Backend & Distributed Platforms, AI Infrastructure, ML Systems

Houston, Texas, USA | +1 832 450 4989 | hasibulamin.cse@gmail.com | LinkedIn | GitHub | Hugging Face | Google Scholar

Software Engineer with 10 years of experience building distributed backend systems, cloud-native platforms, and ML-ready data infrastructure. Experienced in **Kafka**, **microservices**, **Docker/Kubernetes**, **CDC**, **CQRS**, high-throughput APIs, secure enterprise APIs, **DRI/live-site ownership**, and production reliability across logistics and marketplace platforms. Applied AI/ML work spans **continual learning**, **LLM fine-tuning**, **RAG/LLM tooling**, NLP, and multimodal VLM evaluation. Combines production systems ownership with AI research to build reliable infrastructure for analytics, ML, and AI-enabled products.

Core Technical Skills

AI / ML Systems: PyTorch, TensorFlow, Model Training, Fine-Tuning, Deep Learning, Continual Learning, NLP, Hugging Face, BERT, LLMs, RAG, PEFT/LoRA/QLoRA, VLM/Video-QA Evaluation
Distributed Systems & Cloud: Kafka, Event-Driven Architecture, CDC, Microservices, CQRS, High-Throughput APIs, AWS, Azure, Docker, Kubernetes, Terraform/Terragrunt, Bicep, Infrastructure as Code
Data & Reliability: PostgreSQL, Oracle, SQL Server, MongoDB, Redis, Materialized Views, Read Replicas, OpenTelemetry, Distributed Tracing, Structured Logging, RBAC
Languages & Frameworks: Python, C#, F#, SQL, JavaScript, C++, ASP.NET/MVC, Entity Framework, Java Spring, GO

Experience

Application Developer (Backend & Distributed Systems)

Jul 2023 – Present

Crane Worldwide Logistics LLC

Houston, TX — Global 3PL logistics provider

- Architected a multi-region, fault-tolerant **Kafka-based microservices** platform in F# with idempotent message handling, retry policies, and **DLQ-backed** recovery to improve reliability across distributed analytical and operational workflows.
- Designed real-time **CDC pipelines using Kafka** to synchronize **tens of millions of production records** under a **30-second max-latency SLA**.
- Implemented CQRS-based Oracle-to-PostgreSQL read replicas with materialized views, reducing invoice-validation response time and CPU utilization by **15–18%**.
- Owned and maintained a mission-critical billing workflow with a **P99 SLO**, delivering **99% of requests within 90 seconds** through Kafka synchronization, document validation, PDF merge processing, and S3 delivery.
- Served as **DRI** for billing and CDC systems, owning **technical decisions**, **cross-team incident coordination**, RCA follow-up, **OpenTelemetry** tracing, **structured logging**, and runbooks; diagnosed most production incidents within **30–60 minutes**.
- Designed secure customer-facing transportation and warehouse APIs using ASP.NET, Entity Framework, and least-privilege RBAC across AWS, Azure, and database layers for enterprise access control and compliance readiness.
- Architected a **human-in-the-loop AI agent workflow** using prompt and context engineering to automate significant portions of the SDLC — ingests Jira tickets, generates code and unit-test suggestions, conducts automated code review, and manages downstream event streaming via Kafka, with human approval required to merge all changes to production.
- Led backend features end-to-end from requirements through Azure/AWS deployment**; coordinated integration dependencies, authored design docs, reviewed code, interviewed candidates, and **advised senior engineers on architecture and API design standards**.

Independent AI/ML Research Collaborator

Late 2024 – Present

Independent AI/ML research conducted alongside full-time engineering role with multi-institution co-authors

- Built reproducible **continual learning training pipelines in PyTorch**, including ablation studies, replay-buffer variants, and inter-cluster regularization methods for evaluating catastrophic forgetting across sequential learning tasks.
- Evaluated **QLoRA fine-tuning** of open-weight LLMs for medical question answering, measuring response quality across MedQuAD (IEEE CICN 2026).
- Contributed to **VLM hallucination-mitigation** research using training-free inference-time intervention and evaluation across **POPE**, **CHAIR**, **MME**, **HallusionBench**, and **AMBER**.
- Audited detector-based grounding metrics for **video-QA reasoning**, identifying vocabulary confounds and proposing controls for more reliable multimodal evaluation.

Graduate Research Assistant (Applied ML & Systems)

2021 – 2023

University of Houston (Das Lab & CBL)

Houston, TX — During M.S. Computer Science coursework

- Built a **GPU-accelerated ML pipeline** for CT image acquisition and reconstruction, reducing end-to-end processing time by **35–40%**.
- Applied unsupervised learning methods including GMM, K-Means, and Fuzzy C-Means for biomedical tumor segmentation and analysis on GPU-enabled compute infrastructure.
- Built an **R&D conversational AI prototype** using Rasa, NVIDIA NeMo, Docker, AWS, and Google Geolocation APIs, gaining hands-on experience with containerized AI services, dialogue orchestration, and cloud deployment workflows.

Lead Software Engineer

Sep 2019 – Jul 2021

Truck Lagbe Ltd. (Startup)

Dhaka, Bangladesh — Digital freight marketplace for on-demand truck hiring and logistics workflows

- Led and mentored 12 engineers and owned backend architecture for a two-sided freight marketplace serving **400K+ daily requests**, **5,000+ concurrent users** with 45–80 ms average API latency.
- Designed high-availability database architecture with **CQRS-style** read/write separation, read models, replication, **trip-age partitioning**, and nightly recovery workflows to reduce primary database load as volume scaled.
- Engineered an **O(N log N) multilingual geolocation search and spatial matching** engine using fuzzy matching, reverse geocoding, hierarchical search, Redis-to-MongoDB caching, and early-exit pruning; achieved 95% accuracy and reduced mapping API calls by **80–85%**.

- Built **A/B-style rollout and performance testing** for mobile dashboard/API variants, comparing app load time and trip-success signals across bid-only vs. suggested price-range pricing flows before broader release.
- Developed a multi-factor **dynamic pricing engine using demand patterns**, seasonality, tolls, weather, road conditions, and **Haversine-based** distance calculations, improving completed bookings by **23%** and monthly revenue by **20–25%**.
- Developed payment gateway integrations in Java Spring, improving mobile transaction processing and payment workflow reliability.
- Modernized infrastructure by migrating production workloads to AWS, improving performance by **20%**; introduced GitLab/Kubernetes **CI/CD pipelines**, reducing deployment time from 2–3 hours to 20–30 minutes.

Associate Senior Programmer

Feb 2017 – Aug 2019

CSL Software Resources Ltd.

Dhaka, Bangladesh

- Built reusable, runtime-configurable security components in C#, ASP.NET, C++, SQL, and stored procedures, enabling dynamic role-, field-, and column-level permissions across enterprise modules.
- Developed **configurable backend infrastructure** where stored procedures, UI visibility, and authorization behavior were database-driven, reducing development and maintenance effort by 10%.
- Built inventory and procurement ERP systems adopted by 400+ companies and a Point-of-Sale platform used across 250+ retail outlets; refactored APIs and SQL queries, improving dashboard response time by 35%.

Software Engineer

Jan 2015 – Jan 2016

GOBD (Startup)

Dhaka, Bangladesh

- Implemented **route-optimization** and real-time tracking algorithms merging live GPS streams, manual intersection reports, and third-party traffic feeds for consumer navigation features.

Publications & Preprints

- **WACV 2027 (Under Review)**: Residual Stream Rebalancing: Training-Free Hallucination Mitigation in Vision-Language Models ; When Detector-Based Grounding Metrics Measure Vocabulary.
- **IEEE CICN 2026**: Boundary-Aware Deep Learning Framework for Breast Cancer Image Segmentation [IEEE Xplore]; Multilingual Financial Fraud Detection Using ML and Transformer Models [IEEE Xplore]; Evaluating PEFT of Open-Weight LLMs for Medical QA [IEEE Xplore].
- **Preprints**: Continual Learning for Adaptive AI Systems [arXiv]; Detecting Conspiracy Theory Against COVID-19 Vaccines [arXiv]; public Hugging Face dataset.

Professional Service

- Technical Program Committee (TPC) Member — IEEE CICN 2026 — Peer Reviewer — PLOS ONE (AI & ML Track)

Education

- **M.S. Computer Science** — University of Houston, 2023 | Coursework: Machine Learning, Adv. Machine Learning, Deep Learning, NLP, Cloud Computing, Digital Image Processing
- **B.S. Software Engineering** — American International University-Bangladesh (AIUB), 2016 | Coursework: Artificial Intelligence, Data Mining, Algorithms, Data Structures, Object-Oriented Programming, Java

Certifications

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** — LLMs, RAG, Semantic Search, Vector Database, LangChain
- **Architecting on AWS** — Corporate training delivered by AWS Dallas/Houston team through Crane Worldwide Logistics
- **Foundations of AI and Machine Learning** — Microsoft / Coursera